

arXiv:2608.10719 — Multiwavelength evidence for off-nuclear IMBH accretion via gravitational focusing.



Background

Direct observational evidence for gravitational-wake accretion onto wandering black holes has remained elusive.

This paper reports a 35,000-solar-mass wandering black hole accreting through a gravitational wake. Multi-epoch spectroscopy reveals upstream and downstream gas components and rapid broad-line variability.

- Broad lines, compact continuum, and variability confirm a 35,000-solar-mass black hole.
- Three gas components trace upstream flow, downstream wake, and optically thick absorbers.
- Absorbers within the capture radius drive rapidly changing-look broad-line emission.

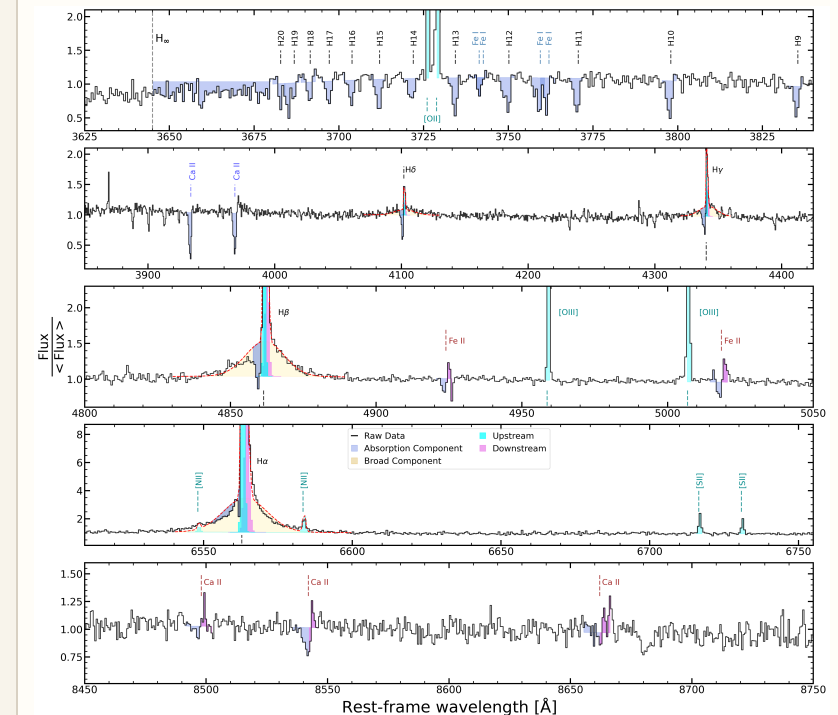


Figure 3 | X-Shooter spectrum decomposes blueshifted absorption, narrow emission, and broad Balmer lines.